SDC HACKATHON 2026

Phase 2: Final Project & Prototype Report

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1 November 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Final.pdf" together with your pitch deck and a working deployment link. Answers must comply with the word limits stated for each open-ended question. Section 3 requires concrete, verifiable evidence. Unsupported claims will not be credited by judges.

1. Team & Project Overview

Team Name	README later
Project / Product Name	Loupe
Final Live Deployment URL	https://halal-supplier-control--dedoh.replit.app
Codebase Repository URL	https://github.com/deettoh/halal-supplier-control
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2. Project Description

[Maximum 250 words]

Loupe is an Orbyte module for Domain 2 (Orbyte Smart SME Challenge: Less Busywork, Better Business) and SDG 8, built by team README later from Xiamen University Malaysia. It checks each delivered ingredient at the receiving door of a halal-certified micro F&B business.

JAKIM's MHMS 2020 requires certified premises to control raw materials through a masterlist and receiving records, including the actual manufacturer (JAKIM, 2020). In a micro SME those records sit in folders, WhatsApp photos and spreadsheets. A familiar brand can arrive as a different product, and a certificate on file can be expired or outside the product's scope, which nobody notices until an audit. Every buyer also chases every supplier for the same certificate.

Staff choose the material and photograph its label. Qwen3-VL-8B-Instruct reads the product, brand, manufacturer and halal mark, and AI is used only for this step. Six Python rules then decide the verdict. Covered or Expiring deliveries are accepted automatically, and every other verdict waits for the Halal Supervisor to accept, reject or hold it with a reason. Each step is logged with the person's name.

A supplier renews once, a registry reviewer verifies it, and every SME that buys from that supplier uses the new certificate. Loupe reports document coverage and gives no halal ruling. Parliament reported 8,105 halal-certified SMEs in July 2024 (Parliament of Malaysia, 2024), and JAKIM had revoked 3,068 certificates for non-compliance by October 2024 (FMT Reporters, 2024).

3. Upgraded Product & Technical Execution

3.1 Final System Architecture

Provide a detailed description (and, if possible, an attached technical diagram) of your fully built system. Explain how your application integrates GenAI APIs, database structures, and backend/frontend frameworks into one unified solution.

Loupe runs as one Replit Autoscale deployment and one RunPod Serverless endpoint. The system block diagram is in **Appendix A**.

• Frontend: React 19 with Vite and Tailwind CSS, built to static files, with screens for receiving staff, supervisors, suppliers and the registry reviewer.

• Backend: FastAPI (Python), which serves the built frontend and the JSON API and checks the caller's person, role and organisation on each route. These request headers simulate identity, and real authentication is a release requirement.

• Database: PostgreSQL on Replit through SQLAlchemy. Certificates are a shared registry with no SME column. Materials, deliveries and the activity log each belong to one SME, and queries filter on it.

• GenAI: Qwen3-VL-8B-Instruct on a RunPod Serverless endpoint with the vLLM worker, called through the OpenAI Python SDK at temperature 0 with a JSON schema for the output. It reads delivery labels and pre-fills the renewal form from certificate photos.

• Decision logic: six Python rules, with rapidfuzz for brand and manufacturer matching. They check that the reading is complete, that it matches the approved material, that the certificate covers the product and hasn't expired, that the issuer is recognised and that a halal mark is visible. R6 checks that a mark is visible, not that it is authentic. Each SME has its own recognised issuers, match threshold and expiry warning days in configuration.

• Hosting: one Replit Autoscale deployment, with the RunPod key in Replit Secrets.

One POST /api/deliveries saves the photo, calls the model, runs the rules and writes the delivery and a log row. The supervisor's decision is a second request, checked for the supervisor role before anything is written. A renewal is a supplier POST /api/certificates that waits as pending until a reviewer request makes it live.

Sponsor resources: yes, both Replit and RunPod are used. Replit Autoscale hosts the whole app: FastAPI serves the React screens and the API, Replit PostgreSQL stores the supplier records, certificates, delivery photos, decisions and activity log, and Replit Secrets hold the RunPod key. RunPod Serverless runs Qwen3-VL-8B-Instruct on a 24 GB GPU. It scales to zero between checks, so cost follows use.

3.2 Implementation & Hackathon Deliverables

Describe what was successfully built, integrated, and deployed during the 48-hour physical sprint. Highlight any specific problem-solving steps your team took to pivot or improve the project from the Phase 1 preliminary design.

[Maximum 300 words]

We identified 4 gaps: self-attested renewals, an unused halal-mark field, logs that recorded only roles and an eight-image evaluation. In the upgrading round (28 to 30 October) we addressed all four. Renewals now wait for a registry reviewer, rule R6 sends a missing mark to the supervisor, log rows name the person, item and verdict, and 24 evaluation cases were added. We also found that a delivery labelled Cocoa Liquor could pass as Cocoa Powder when the brand matched, so R2 now compares the product as well.

During the physical sprint on 31 October we built and merged these items:

• the frontend connection for Hold and certificate photo pre-fill

• one renewal request per SME and certificate in the expiry check

• fuller audit logs, with seeded history removed so every row comes from a real action

• renewal flags on the supplier's certificates, and a simplified mobile view for staff

• photo capture from the camera or saved photos, and flour labels in the supervisor queue

• the delivery check and review timing procedure, with its results

The finals evaluation first scored 66 of 72 verdicts. We revised the prompt to keep customer names out of the manufacturer field and reran it for 72 of 72.

Two Phase 1 plans changed. WhatsApp submission and the Verify Halal cross-check moved to the roadmap so the team could fix correctness and trust first. Expiry monitoring runs when the reviewer starts it, and a daily schedule isn't built yet.

4. Environmental & SDG Impact Verification

Validate the real-world impact of your built solution. Show concrete data that validate your environmental or social value proposition, compared against the baseline and KPI defined in previous phase submission.

*The data and evidence you provide in this section will be used to assess **SDG Impact, Technical Execution and Innovation scores.** Vague claims or missing verification will weaken your score across all three categories.*

Metric Category	**Your Response**
Target SDG Metric Achievement *Direct comparison between your Phase 1 baseline and your Phase 2 finalized prototype numbers* *e.g., "Achieved a 24.5% reduction in computational execution overhead against the 20% initial target."*	Delivery check: median 15.5 s in Loupe against 92 s by hand, which is 83.2% less time. A team member timed Loupe on 32 synthetic labels in three rounds (96 checks), and timed the manual check once on the same 32 labels. Every check by both methods reached the expected result. Steps for staff fall from 8 to 3. Supervisor review: median 31 s in Loupe against 320 s by hand, which is 90.3% less, over the 22 labels that need review (66 reviews in Loupe, 22 by hand). Against Phase 1: the target was at least 70% less time per check, about 25 s or less against the 82 s baseline. The result meets it both against the 92 s manual time on the same 32 labels (83.2%) and against the 82 s baseline (81.1%). Phase 1 also asked for labels held out from prompt tuning, and these 32 aren't held out, since the prompt was revised on the finals cases. Detection: across the evaluation runs and the 8 Phase 1 cases, all 26 wrong-product and substitution attempts were flagged, none of the 26 clean attempts was flagged, and both Phase 1 cases with an expired or out-of-scope certificate were flagged. The Phase 1 targets were 90%, at most 5% false flags and 100%. Reading accuracy: on 24 AI-generated label cases run three times with the demo cache disabled, 72 of 72 verdicts matched their targets, and every case kept the same verdict across the three runs (24 of 24). Readable text matched in

Metric Category	Your Response
	204 of 210 fields as written (97.14%) and 210 of 210 after normalisation, and median extraction took 3.14 s (p95 3.50 s) (README later, 2026a). This supports SDG 8.2 (productivity through technological upgrading) and SDG 8.3 (support for micro and small enterprises) (United Nations, n.d.).
Verification Evidence Method *How was this data verified?* *e.g., codebase algorithmic profiling logs, hardware power-meter simulation readings, or database transaction counters.*	Timing: docs/measurements/Timing-delivery.xlsx and Timing-review.xlsx in the repo list every timed case with its start, stop, expected and actual result, following docs/timing-procedure.md. These are timings on synthetic deliveries by the team, not a field study. Evaluation: tests/eval/finals/results/manufacturer-prompt/ keeps case-level predictions, three run reports, a combined summary and configuration hashes, so each total traces back to individual cases. The 24 cases are variants of 8 packaging designs, and targets were revised during development. With no untouched holdout and no SME pilot, this shows consistency on a development set, not accuracy on unseen deliveries. Code: tests/test_expiry_check.py covers expiry windows, repeated requests, a later request for a second SME, rollback after a failed log write and concurrent reviewer clicks. tests/test_registry_renewals.py covers pending renewals, reviewer decisions and named logs. The server refuses a decision from any role except supervisor, and judges can reject a delivery and see the log row it writes.
Socio-Economic / Civic Value *How does your deployed application translate into a scalable, real-world solution capable of being supported by civic or corporate partners?*	Receiving staff get a clear next step, supervisors attach a person, reason and time to each decision, and suppliers keep one certificate record for all their buyers. In the demo, one verified renewal reached 9 items across 2 SMEs, while each SME kept its own masterlist, settings and decisions. Each delivery check takes 76.5 s less than by hand (83.2%), and each flagged delivery takes almost 5 minutes less to decide (90.3%). The main value is protecting the

Metric Category	Your Response
	halal certificate. MHMS 2020 requires raw-material controls and receiving records that identify the supplier, manufacturer and supporting evidence (JAKIM, 2020), and an audit finding that isn't corrected in time can lead to suspension (FMT Reporters, 2024). For a halal bakery, losing the certificate can mean losing its Muslim customers and the buyers that require halal certification. Loupe catches a wrong delivery or a lapsed supplier certificate at the door, before the audit, and leaves an audit-ready record of every check. The 8,105 certified SMEs (Parliament of Malaysia, 2024) and 3,068 revoked certificates (FMT Reporters, 2024) show the size of the ecosystem and the cost of losing certification. They don't count affected businesses or losses Loupe has prevented. JAKIM's 2025 MyHALALINGREDIENTS initiative records and assesses raw materials to reduce repeated documentation in certification (JAKIM, 2025). Loupe complements it by checking delivered packages against approved evidence, with no government integration or endorsement. Privacy under PDPA: the only personal data is staff names and roles, each SME sees only its own records, and photos go only to our own model endpoint. A real deployment would add staff consent at onboarding, a retention period tied to audit needs and a data processing agreement with the GPU provider.
Deployment & Commercial Feasibility *e.g., modeled as an API service plugin designed for easy integration into existing corporate or institutional systems.*	Loupe is modelled as a usage-priced module for the Orbyte hub. A second business is a setup change, where it adds its masterlist, people and settings, and Crescent Caterer already runs on the same code with 45 warning days against Sunrise Bakery's 60. Login, organisations and supplier records are simulated, with assumed Orbyte interfaces.

Metric Category	Your Response
	Proposed price: RM0.50 per check, with suppliers joining free, so 130 checks a month costs RM65. That makes Loupe low-cost insurance against losing the certificate. We haven't tested the price with customers, and the pilot measures willingness to pay directly. Modelled computing cost per check, at listed rates: RM0.34 at 100 checks a month, RM0.31 at 1,000 and RM0.13 at 10,000, which leaves a margin over computing cost of 32%, 38% and 74%. The volumes are illustrative, and the database, staff, support, payment fees and tax aren't included, so this isn't profit. Before release: real authentication, a tenant-isolation review, retention controls and monitoring.

4.1 Mathematical Proof / Code Logic Snippet (Optional, but holds bonus marks)

Paste the specific code section, mathematical formula, simulation log output, or calculation logic that proves the impact claims made above.

This proof is graded across three categories at once: ***Impact and Feasibility*** *(does the math support your claim),* ***Technical Execution*** *(does it reflect what you actually built), as well as* ***Innovation and Creativity*** *(is the approach original).] Do not leave this section generic or copied from Phase 1.*

Proof 1. Time saved, from the medians in docs/measurements/:

```
delivery = (92.0 - 15.5) / 92.0 * 100     # 83.2% less
review = (320.0 - 31.0) / 320.0 * 100     # 90.3% less
vs_phase1 = (82.0 - 15.5) / 82.0 * 100    # 81.1% less
```

Proof 2. Expiry requests follow each SME's own warning window. Simplified from expiry_check in src/backend/services.py:

```
days_remaining = (certificate.expiry_date - today).days
for tenant in linked_smes(certificate):
    qualifies = 0 <= days_remaining <= settings(tenant).expiry_warning_days
    if qualifies and not already_requested(certificate, tenant):
        log_renewal_request(certificate, tenant)
```

At 60 days left, Sunrise Bakery (60-day window) gets a request and Crescent Caterer (45-day window) doesn't yet. A repeat run creates no duplicate. At 45 days, Crescent gets its first request and Sunrise's isn't repeated. Row locks and a database constraint prevent duplicates, as tested by test_sme_windows_and_repeat_clicks, test_each_sme_gets_its_request_when_its_window_starts and test_concurrent_reviewer_clicks_create_one_request.

Proof 3. A renewal changes coverage only after review. The supplier's replacement is saved as pending and the active certificate stays in use. When the reviewer verifies it, the replacement becomes active and the old one superseded, both linked SMEs' materials use the new expiry, and each SME gets its own submission and decision entries in its log (test_pending_renewal_leaves_both_smes_unchanged, test_reviewer_decision_updates_coverage_and_named_logs).

Proof 4. Evaluation arithmetic, from tests/eval/finals/results/manufacturer-prompt/summary.md:

```
24 cases x 3 runs = 72 attempts
72 matching verdicts / 72 attempts x 100 = 100%
# 72 repeated attempts on 24 cases, not 72 independent photos
```

Proof 5. Modelled cost per check, at listed rates:

```
usd_per_s, rm_per_usd = 0.00037, 4.08
cold = (137 + 1.5 + 60) * usd_per_s * rm_per_usd    # about RM0.30
warm_month = 9 * 3600 * 26 * usd_per_s * rm_per_usd  # about RM1,272
breakeven = warm_month / cold   # about 4,300 checks a month
```

5. Market Viability & Future Scalability Plan

SDC aims to encourage solutions that transition from innovations into market-ready applications. If given access to corporate funding, industry grants, or partnership opportunities (e.g., through SDC's Rotary network or joint research initiatives at XMUM), how would your team scale this prototype to process real-world data at volume?

[Maximum 250 words]

Funding would first pay for a pilot with one halal-certified premise, then a second buyer that shares a supplier, which tests onboarding effort and the shared registry together. The pilot would record task time, discrepancy handling, reviewer decisions, onboarding effort, repeat use, cost and willingness to pay. It would repeat the timing on photographed deliveries at the premise, with labels held out from prompt tuning as the Phase 1 target asked. A larger held-out label set is a natural joint research project with XMUM.

For the product, the next steps are WhatsApp submission, authorised cross-checks against certification services such as Verify Halal, sign-in through Orbyte and a daily expiry check with supplier reminders.

At low volume each check wakes the model, at about RM0.31 a check. Above about 4,300 checks a month, keeping one worker warm for working hours costs about RM1,280 a month and becomes cheaper, reaching about RM0.13 a check at 10,000 checks, which is only about 10 minutes of GPU work a day. These figures are modelled from listed rates.

The registry grows with scale, since each supplier who joins makes Loupe more useful to every buyer. We would start with bakeries and caterers on the Orbyte hub, then restaurants and central kitchens, and Rotary's business network could help reach the first pilot premises.

6. Final Submission Checklist

Tick all boxes that apply before submitting.

☑ The application is fully compiled, live, and accessible via the deployment link provided in Section 1.

☑ The GitHub (or equivalent) repository contains a clean, well-documented README with installation instructions.

☑ The GitHub (or equivalent) repository shows the history of incremental code updates made throughout the official live hacking window.

☑ The final pitch deck has been exported and saved as a flattened PDF file.

☑ All fields in this template have been completed and word limits respected.

7. IP & Originality Declaration

7.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No ☐ No change from Phase 1 declaration

If yes, briefly describe what was pre-existing and which team member(s) owned it:

7.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No ☐ No change from Phase 1 declaration

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Qwen3-VL-8B-Instruct	Model	Apache-2.0
vLLM	Code (model server)	Apache-2.0
FastAPI, Uvicorn	Code	MIT; BSD-3-Clause
SQLAlchemy, psycopg	Code	MIT; LGPL-3.0
Pydantic, python-multipart	Code	MIT; Apache-2.0
OpenAI Python SDK, certifi, tzdata	Code	Apache-2.0; MPL-2.0; Apache-2.0
rapidfuzz, Pillow	Code	MIT; MIT-CMU
React, Vite, Tailwind CSS, TanStack Query	Code	MIT
wouter, react-aria-	Code	Unlicense; Apache-2.0; ISC

components, lucide-react

three, @react-three/fiber, @react-three/drei	Code	MIT
GSAP, @gsap/react	Code	GSAP Standard No Charge License
Public Sans and Barlow Semi Condensed fonts	Font	OFL-1.1

7.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☑ No ☐ No change from Phase 1 declaration

If yes, please describe the nature of the agreement and which member it concerns:

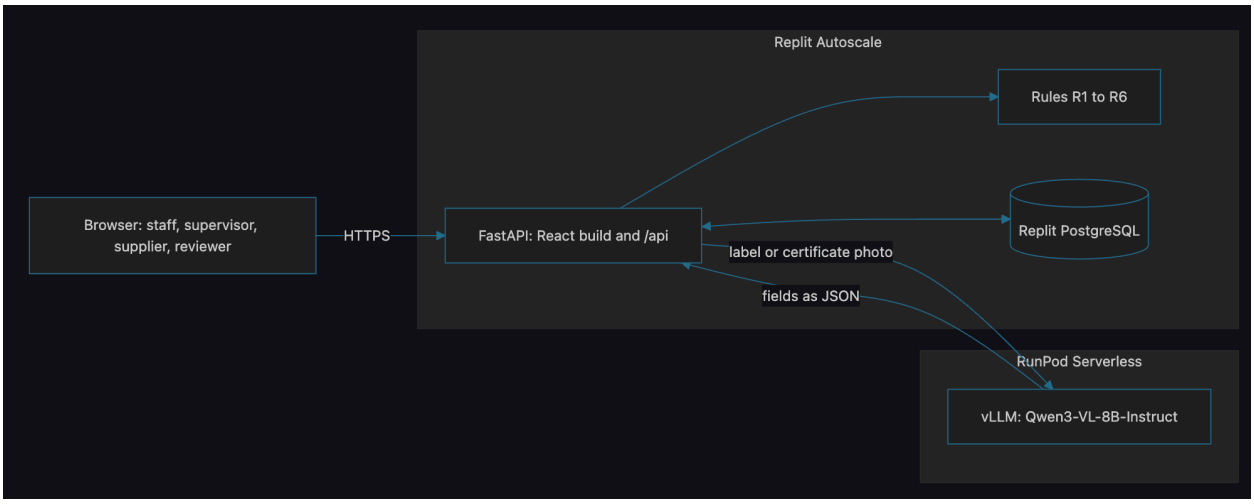
8. Declaration

By submitting this report, the team leader confirms that the deployed application and data presented above are the original work of the team, completed within the official SDC 2026 hackathon period.



31/10/2026

Appendix A


Replit Autoscale
Rules R1 to R6
Browser: staff, supervisor, supplier, reviewer
HTTPS
FastAPI: React build and /api
Replit PostgreSQL
label or certificate photo
fields as JSON
RunPod Serverless
vLLM: Qwen3-VL-8B-Instruct

References

FMT Reporters. (2024, October 12). Jakim tarik balik 3,068 sijil halal. Free Malaysia Today. https://www.freemalaysiatoday.com/category/bahasa/tempatan/2024/10/12/jakim-tarik-balik-3068-sijil-halal

JAKIM (Department of Islamic Development Malaysia). (2020). Sistem pengurusan halal Malaysia 2020 [Malaysian Halal Management System 2020]. https://halal.myeg.com.my/assets/files/MHMS_2020.pdf

JAKIM (Department of Islamic Development Malaysia). (2025, August 14). Kenyataan media JAKIM berkenaan pelaksanaan MyHALALINGREDIENTS [Media statement on the implementation of MyHALALINGREDIENTS]. https://www.islam.gov.my/en/media-statement/4798-kenyataan-media-jabatan-kemajuan-islam-malaysia-jakim-berkenaan-pelaksanaan-myhalalingredients

Parliament of Malaysia. (2024, July 18). Penyata rasmi Dewan Rakyat, 18 Julai 2024 [Dewan Rakyat proceedings, 18 July 2024]. https://hansard.parlimen.gov.my/hansard/dewan-rakyat/2024-07-18

README later. (2026a). Finals three-run evaluation: manufacturer prompt [Evaluation report]. https://github.com/deettoh/halal-supplier-control/blob/main/tests/eval/finals/results/manufacturer-prompt/summary.md

United Nations. (n.d.). Goal 8. United Nations Department of Economic and Social Affairs. https://sdgs.un.org/goals/goal8